

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1764

Mickens 345 kV
July 22, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1764. The Interconnection Customer proposes a 274.0 MW battery energy storage facility interconnecting at the Mickens 345 kV substation in the Virginia area. The request was received on May 29, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 587 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,067,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Morrell Substation / Old Church 138 kV
Contingency: Skaggs Island Jct 2 / Chancellor Row 345 kV
Loading: 123.4% (Rating: 232 MVA)
DFAX: 5.3%
Pre-project loading: 117.1%
2. Facility: Tippecanoe / Old Church 138 kV line
Contingency: Skaggs Island Junction 2 / Chancellor Row 345 kV
Loading: 120.5% (Rating: 266 MVA)
DFAX: 5.3%
Pre-project loading: 115.1%
3. Facility: Skaggs Island Junction 2 / Tippecanoe 138 kV
Contingency: Skaggs Island Junction 2 / Chancellor Row 345 kV L
Loading: 118.6% (Rating: 295 MVA)
DFAX: 5.3%
Pre-project loading: 113.7%
4. Facility: Hickory / Elmo 138 kV
Contingency: Mira Vista / Yarnen 345 kV
Loading: 108.3% (Rating: 323 MVA)
DFAX: 11.6%
Pre-project loading: 98.5%
5. Facility: Hospital Distrito 7003 / Hardy Reit 138 kV
Contingency: Mitchell Dam / Skaggs Island Junction 2 345 kV
Loading: 105.0% (Rating: 220 MVA)
DFAX: 5.0%
Pre-project loading: 98.8%

6. Facility: Bear Island 138/1 kV Transformer
 Contingency: Bear Island / Chancellor Row 345 kV L
 Loading: 100.4% (Rating: 513 MVA)
 DFAX: 7.2%
 Pre-project loading: 96.5%
7. Facility: Leavitt / Birchwood Substation 345 kV line
 Contingency: Dunn Siding Substation / Bear Island 345 kV
 Loading: 100.3% (Rating: 3,527 MVA)
 DFAX: 22.7%
 Pre-project loading: 98.6%
8. Facility: Mitchell Dam / Haggin 345 kV
 Contingency: Bear / Mecklenburg Power Sta 345 kV
 Loading: 100.3% (Rating: 1,658 MVA)
 DFAX: 21.5%
 Pre-project loading: 96.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Morrell Substation / Old Church 138 kV	Rebuild line with higher-capacity conductor	\$25,476,000	5% (\$1,254,000)
Old Church / Tippecanoe 138 kV	Rebuild line with higher-capacity conductor	\$5,417,000	5% (\$267,000)
Tippecanoe Substation / Skaggs Island Jct 2 138 kV	Rebuild line with higher-capacity conductor	\$4,334,000	5% (\$213,000)
Hickory / Elmo 138 kV	Reconductor line	\$785,000	5% (\$43,000)
Hardy Reit / Hospital Distrito 7003 138 kV Line	Reconductor line	\$4,349,000	8% (\$356,000)
Bear Island 138/1 kV Transformer	Replace transformer with higher-rated unit	\$3,345,000	3% (\$95,000)
Leavitt / Birchwood Substation 345 kV line	Reconductor line	\$57,618,000	3% (\$1,632,000)
Mitchell Dam / Haggin 345 kV	Reconductor line	\$6,363,000	3% (\$207,000)
	Total allocated to project		\$4,067,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.